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M Pozulp, T Haut

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A Proof of the Asymptotic Variance of Path Length Estimators for Single-Collision Monte Carlo Source Iteration in the Thick Diffusion Limit

Michael Pozulp[a,*], Terry Haut[b]

[a] *Applications, Simulations, and Quality Division, Lawrence Livermore National Laboratory, Livermore, California 94550*
[b] *Center for Applied Scientific Computing, Lawrence Livermore National Laboratory, Livermore, California 94550*

Abstract

We prove a theorem relating the variance of path length estimators for single-collision Monte Carlo source iteration to a parameter that becomes infinitesimally small in an important physical regime arising in radiative transfer. In our usage, "single-collision Monte Carlo source iteration" refers to Monte Carlo Boltzmann transport methods in which each Monte Carlo particle history includes no more than a single collision, and the physics of multiple scattering is modeled by lagging the scattering source term and iterating until this term converges.

Our theorem can be used to construct variance reduction techniques which improve the order of the estimator variance. This enables calculations that would otherwise require impractically large sample sizes to achieve practical estimator uncertainties. We believe this is the first postulation of a theorem relating estimator variance to a limiting case parameter for single-collision Monte Carlo source iteration, and the first proof of such a theorem.

We illustrate the theorem's value with an example in which the authors of a transport method used the theorem to design a variance reduction technique that improved the uncertainty of their solution by a factor of about 500 for a proxy problem from radiative transfer that contains both optically-thick and optically-thin material.

Keywords: Monte Carlo, variance reduction, thick diffusion limit

1. Introduction

It is not uncommon to encounter substantial statistical variation when using Monte Carlo to compute the solution of a Boltzmann transport equation. This variation, or noise, can degrade solution quality enough that the Monte Carlo estimators become useless. The Monte Carlo noise can be systematically reduced through sample augmentation, but the sample size required for an acceptable solution estimate may be impractical. The variance of a function of random variables f is,

$$\mathrm{Var}[f] = E[f^2] - (E[f])^2 \,, \tag{1}$$

where $E[\cdot]$ is the expectation, or expected value. Equation (1) shows that computing the variance requires the definition of f and the expectation, which is the first moment of f about the origin over its support.

Suppose that we want to use Monte Carlo to estimate some quantity F. The Monte Carlo simulation procedure can be viewed as defining f such that $E[f] = F$, then computing the Monte Carlo estimator $\hat{F}$ that approximates F by generating random variates of f and averaging the variates,

$$F \approx \hat{F} = \frac{1}{N} \sum_{n=1}^{N} f^{(n)} \,, \tag{2}$$

*Corresponding author (pozulp1@llnl.gov)

where $f^{(n)}$ is the nth random variate. Using Equation (2) and assuming that the $f^{(n)}$ are independent and identically distributed, the variance of our estimator is,

$$\mathrm{Var}[\hat{F}] = \frac{\mathrm{Var}[f]}{N}\,, \tag{3}$$

from which we see how sample augmentation (increasing N) and variance reduction (decreasing $\mathrm{Var}[f]$) reduce noise.

Asymptotic analysis is a technique that can be used to study how a numerical method behaves as an asymptotic analysis parameter approaches a particular limit. If a calculation which exercises the physical regime associated with the limiting case produces a noisy Monte Carlo estimator, then the estimator variance may have an undesirable dependence on the asymptotic analysis parameter. In the worst case, the estimator variance approaches infinity as the asymptotic analysis parameter approaches its limit. In the best case, the estimator variance approaches zero. If the limiting case is important to the application, then the difference between the best case and the worst case is the difference between a useful Monte Carlo method and a useless one. The theorem that we prove is a useful mathematical tool for this exact scenario.

In that sense, the theorem can be used as a powerful tool for designing variance reduction techniques. For example, the authors of a Monte Carlo method for Boltzmann transport used the theorem to modify their method to construct estimators with variances that approach zero in the limiting case. This led to a substantial improvement: zero variance instead of infinite variance in the limit. We outline their method and results in Section 4. The purpose of this manuscript is to prove the theorem, thereby providing assurance that the empirical variance estimates will agree with the theoretical analytic variances when the number of samples is sufficiently large.

Our proof relies on the expression of a Monte Carlo estimator as an expected value of a function of random variables. This is a perspective which often appears in descriptions of Monte Carlo mathematics for Boltzmann transport, such as the textbooks by Spanier and Gelbard [1] and Lux and Koblinger [2]. Both texts present the collision density equation, the Fredholm integral equation of the second kind, and the Neumann series expansion, which the authors use in their analysis of multi-collision Monte Carlo. Our presentation is simplified by the restriction that the Monte Carlo particles can collide no more than once, though we do not impose any limit on the number of collisions undergone by physical particles. We achieve this by handling the physics of multiple collisions using source iteration, which lags the scattering source in order to avoid simulating scattering events in the Monte Carlo particle histories. Lagging the scattering source is uncommon in Monte Carlo methods but is essential for deterministic methods, where source iteration avoids forming and storing the transport operator [3].

The theorem that we prove relates the variance defined in Equation (3) to an asymptotic analysis parameter which becomes infinitesimally small in an important physical regime that arises in radiative transfer. We describe the regime in Section 2, as well as some background information on the linear Boltzmann transport equation that we consider, before ending the section with a statement of the theorem. In Section 3, we prove the theorem, discuss some limitations of the proof, and state a corollary together with its proof. In Section 4, we provide an example of how the theorem can be applied to convert a method with a variance that limits to infinity into one that limits to zero. This resulted in an improvement in the uncertainty of a Monte Carlo estimator of the angle integrated intensity by a factor of about 500 for a proxy problem from radiative transfer that contains both optically-thick and optically-thin material. We conclude with a summary in Section 5.

2. Statement of the Theorem

A statement of the theorem requires some background information on the Boltzmann transport problem under consideration. We provide this background information first, before finishing this section with the statement of the theorem. The Boltzmann transport problem that we consider is,

$$\mathbf{\Omega} \cdot \nabla \psi + \sigma_t \psi = Q\,, \tag{4a}$$

where $\mathbf{\Omega} \in \mathbb{R}^3$ is the unit direction vector, ψ is the radiation intensity, σ_t is the total material opacity, and Q is the total source function. Equation (4a) is subject to an inflow boundary condition, which defines the radiation intensity on the domain boundary in the hemisphere of directions that point into the domain at a given location on the domain boundary surface,

$$\psi(\mathbf{x}, \mathbf{\Omega}) = \psi_{\text{inc}}(\mathbf{x}, \mathbf{\Omega}) \,, \quad \mathbf{x} \in \partial\mathcal{D} \text{ and } \mathbf{\Omega} \cdot \mathbf{n} < 0 \,, \tag{4b}$$

where $\mathbf{x} \in \mathbb{R}^3$ is the spatial position coordinate, ψ_{inc} is the inflow source function, $\partial\mathcal{D}$ is the domain boundary, and $\mathbf{n}$ is the outward facing unit normal vector on the domain boundary surface. To model a problem in which photons are scattered and emitted by electronic interactions with the material, we could define Q to be,

$$Q = \frac{\sigma_s}{4\pi} \int_{\mathbb{S}^2} \psi \, \mathrm{d}\Omega' + q \,, \tag{5}$$

where σ_s is the scattering opacity, $\mathbb{S}^2$ is the unit sphere, and q is an arbitrary source function with which to account for other ways that photons may enter phase space, such as photon emission from electronic de-excitations in the material.

A common approach for the Monte Carlo simulation of the Boltzmann transport Equation (4a) with Q defined by the total source function Equation (5) involves simulating scattering events. In this approach, when a simulation particle undergoes a collision identified as a scattering event, it emerges with a new direction and a new frequency, both of which are chosen by pseudo-random sampling of the appropriate distributions. Multiple collisions can occur because scattering events do not terminate the particle history.

An alternative approach is to incorporate the contribution of the scattering source in the weight of the simulation particles. This necessitates an iteration to compute successive approximations of the integral in Equation (5), but it removes the possibility of multiple collisions. Without scattering events, simulation particles can only have at most one collision: a history with a single collision corresponds to a particle that was absorbed in the domain, and a history with no collisions corresponds to a particle that leaked through a vacuum boundary surface. The approach without scattering events is the one that we consider. Additionally, we choose to disregard frequency dependence as well as time dependence in the Boltzmann transport problem that we consider, hence why we write $\psi = \psi(\mathbf{x}, \mathbf{\Omega})$ instead of $\psi = \psi(\mathbf{x}, \mathbf{\Omega}, \nu, t)$.

The limiting case corresponding to the asymptotic analysis that we apply is called the thick diffusion limit (TDL) [4]. It is an optically-thick, highly-scattering transport regime characterized by an asymptotic analysis parameter $\varepsilon \in (0, 1]$. Mathematically, the regime corresponds to a particular scaling of the transport problem data by powers of ε, followed by taking the limit $\varepsilon \to 0$. First, define new variables to represent the scaled problem data,

$$\sigma_t^\varepsilon = \sigma_t / \varepsilon \,, \tag{6a}$$

$$\sigma_a^\varepsilon = \sigma_a \varepsilon \,, \tag{6b}$$

$$q^\varepsilon = q\varepsilon \,, \tag{6c}$$

where $\sigma_a = \sigma_t - \sigma_s$ is the absorption opacity. The unscaled problem data are independent of ε and thus order unity,

$$\sigma_t \sim 1 \,, \tag{7a}$$

$$\sigma_a \sim 1 \,, \tag{7b}$$

$$q \sim 1 \,. \tag{7c}$$

Therefore, the order dependence of the scaled problem data on ε is,

$$\sigma_t^\varepsilon \sim \varepsilon^{-1} \,, \tag{8a}$$

$$\sigma_a^\varepsilon \sim \varepsilon \,, \tag{8b}$$

$$q^\varepsilon \sim \varepsilon \,. \tag{8c}$$

Applying this scaling to the Boltzmann transport problem given by Equation (4a) with boundary condition Equation (4b) produces a new Boltzmann transport problem for ψ^ε,

$$\mathbf{\Omega} \cdot \nabla\psi^\varepsilon + \sigma^\varepsilon\psi^\varepsilon = Q^\varepsilon \,, \tag{9a}$$

$$\psi^\varepsilon(\mathbf{x}, \mathbf{\Omega}) = \psi^\varepsilon_{\text{inc}}(\mathbf{x}, \mathbf{\Omega}) \,, \quad \mathbf{x} \in \partial\mathcal{D} \text{ and } \mathbf{\Omega} \cdot \mathbf{n} < 0 \,, \tag{9b}$$

where we used the symbol σ^ε in place of σ^ε_t for notational simplicity, and where Q^ε is defined as,

$$Q^\varepsilon = \frac{\sigma^\varepsilon_t - \sigma^\varepsilon_a}{4\pi} \int_{\mathbb{S}^2} \psi^\varepsilon \, \mathrm{d}\Omega' + q^\varepsilon \,. \tag{9c}$$

We can now state the theorem which connects the variance of Monte Carlo estimators for the Boltzmann transport Equation (9a) and boundary condition Equation (9b) to the TDL parameter ε. The theorem is as follows:

Theorem 1. *Let $0 < \varepsilon \ll 1$ and let $\mathcal{D} \subset \mathbb{R}^{\dim}$ $(\dim = 1, 2, 3)$ be the spatial domain, $\mathbb{S}^2$ the angular domain, $\partial\mathcal{D}$ the boundary of $\mathcal{D}$, and $\mathbb{S}^2_h$ the unit hemisphere of angles directed into $\mathcal{D}$ at a given point on $\partial\mathcal{D}$. Further, let $Q^\varepsilon : \mathcal{D} \times \mathbb{S}^2 \to \mathbb{R}$, $\psi^\varepsilon_{\text{inc}} : \partial\mathcal{D} \times \mathbb{S}^2_h \to \mathbb{R}$, and $u^\varepsilon : \mathbb{S}^2 \to \mathbb{R}$ have leading terms in ε of $Q^\varepsilon \sim \varepsilon^{\alpha_Q}$, $\psi^\varepsilon_{\text{inc}} \sim \varepsilon^{\alpha_{\text{inc}}}$ and $u^\varepsilon \sim \varepsilon^{\alpha_u}$. Let $\sigma^\varepsilon = \frac{\sigma}{\varepsilon}$ with an otherwise ε-independent parameter $\sigma \in [0, \infty)$. Let ψ^ε solve,*

$$\mathbf{\Omega} \cdot \nabla\psi^\varepsilon + \sigma^\varepsilon\psi^\varepsilon = Q^\varepsilon \,, \tag{10}$$

with boundary data $\psi^\varepsilon_{\text{inc}}$ and $\mathbf{\Omega} \in \mathbb{S}^2$. Given $K \subset \mathcal{D}$, let $F = \frac{1}{\mathrm{vol}(K)} \int_K \int_{\mathbb{S}^2} u^\varepsilon\psi^\varepsilon \, \mathrm{d}\Omega \, \mathrm{d}\mathbf{x}$ with $\mathbf{x} \in \mathcal{D}$ and let $\hat{F}$ be a Monte Carlo path length estimator of F with N interior and M boundary samples. Then there exists a function $G(\varepsilon)$ such that $\mathrm{Var}[\hat{F}] \leq G(\varepsilon)$ where,

$$G(\varepsilon) \sim \varepsilon^{2\alpha_u} \left(\frac{\varepsilon^{2\alpha_Q + 1}}{N} + \frac{\varepsilon^{2\alpha_{\text{inc}} + 1}}{M} \right) \quad \text{as } \varepsilon \to 0 \,. \tag{11}$$

The following corollary identifies three choices of the arbitrary weighting function $u^\varepsilon = u^\varepsilon(\mathbf{\Omega})$, each corresponding to one of the first three angular moments of the intensity. These moments are physical quantities that are widely used in applications of Boltzmann transport, including the application discussed in Section 4. The corollary is as follows:

Corollary 1. *Define the first three angular moments of the intensity by,*

$$\phi^\varepsilon = \int_{\mathbb{S}^2} \psi^\varepsilon(\mathbf{x}, \mathbf{\Omega}) \, \mathrm{d}\Omega \,, \tag{12}$$

$$\boldsymbol{J}^\varepsilon = \int_{\mathbb{S}^2} \mathbf{\Omega} \, \psi^\varepsilon(\mathbf{x}, \mathbf{\Omega}) \, \mathrm{d}\Omega \,, \tag{13}$$

$$\mathbf{P}^\varepsilon = \int_{\mathbb{S}^2} \mathbf{\Omega} \otimes \mathbf{\Omega} \, \psi^\varepsilon(\mathbf{x}, \mathbf{\Omega}) \, \mathrm{d}\Omega \,, \tag{14}$$

where $\otimes$ denotes the tensor (outer) product. Consider the averages of the moments over an arbitrary mesh element $K \subset \mathcal{D}$,

$$\phi^\varepsilon_K = \frac{1}{\mathrm{vol}(K)} \int_K \phi^\varepsilon \, \mathrm{d}\mathbf{x} = \frac{1}{\mathrm{vol}(K)} \int_K \int_{\mathbb{S}^2} \psi^\varepsilon(\mathbf{x}, \mathbf{\Omega}) \, \mathrm{d}\Omega \, \mathrm{d}\mathbf{x} \,, \tag{15}$$

$$\boldsymbol{J}^\varepsilon_K = \frac{1}{\mathrm{vol}(K)} \int_K \boldsymbol{J}^\varepsilon \, \mathrm{d}\mathbf{x} = \frac{1}{\mathrm{vol}(K)} \int_K \int_{\mathbb{S}^2} \mathbf{\Omega} \, \psi^\varepsilon(\mathbf{x}, \mathbf{\Omega}) \, \mathrm{d}\Omega \, \mathrm{d}\mathbf{x} \,, \tag{16}$$

$$\mathbf{P}^\varepsilon_K = \frac{1}{\mathrm{vol}(K)} \int_K \mathbf{P}^\varepsilon \, \mathrm{d}\mathbf{x} = \frac{1}{\mathrm{vol}(K)} \int_K \int_{\mathbb{S}^2} \mathbf{\Omega} \otimes \mathbf{\Omega} \, \psi^\varepsilon(\mathbf{x}, \mathbf{\Omega}) \, \mathrm{d}\Omega \, \mathrm{d}\mathbf{x} \,. \tag{17}$$

Let $\hat{F}_0$, $\hat{F}_1$, and $\hat{F}_2$ denote the Monte Carlo path length estimators of ϕ_K^ε, $\boldsymbol{J}_K^\varepsilon$, and $\mathbf{P}_K^\varepsilon$, respectively, constructed as in Theorem 1 with the following choices of weighting function:

$$u_0(\mathbf{\Omega}) = 1\,, \tag{18}$$
$$\boldsymbol{u_1}(\mathbf{\Omega}) = \mathbf{\Omega}\,, \tag{19}$$
$$\mathbf{u_2}(\mathbf{\Omega}) = \mathbf{\Omega}\otimes\mathbf{\Omega}\,. \tag{20}$$

Then the following hold:

1. *For $u_0 = 1$, the estimator $\hat{F}_0$ is a scalar path length estimator of the element average ϕ_K^ε, and its variance satisfies*
$$\mathrm{Var}[\hat{F}_0] \;\le\; G_0(\varepsilon) \sim \frac{\varepsilon^{2\alpha_Q+1}}{N} + \frac{\varepsilon^{2\alpha_{\mathrm{inc}}+1}}{M} \quad \text{as } \varepsilon\to 0\,. \tag{21}$$

2. *For $\boldsymbol{u_1} = \mathbf{\Omega}$, the estimator $\hat{F}_1$ is a vector-valued path length estimator of the element average $\boldsymbol{J}_K^\varepsilon$, and each component satisfies*
$$\mathrm{Var}[(\hat{F}_1)_\ell] \;\le\; G_{1,\ell}(\varepsilon) \sim \frac{\varepsilon^{2\alpha_Q+1}}{N} + \frac{\varepsilon^{2\alpha_{\mathrm{inc}}+1}}{M} \quad \text{as } \varepsilon\to 0\,, \quad \ell = 1,\ldots,\dim\,. \tag{22}$$

3. *For $\mathbf{u_2} = \mathbf{\Omega}\otimes\mathbf{\Omega}$, the estimator $\hat{F}_2$ is a tensor-valued path length estimator of the element average $\mathbf{P}_K^\varepsilon$, and each component satisfies*
$$\mathrm{Var}[(\hat{F}_2)_{ij}] \;\le\; G_{2,ij}(\varepsilon) \sim \frac{\varepsilon^{2\alpha_Q+1}}{N} + \frac{\varepsilon^{2\alpha_{\mathrm{inc}}+1}}{M} \quad \text{as } \varepsilon\to 0\,, \quad i,j = 1,\ldots,\dim\,. \tag{23}$$

3. Proof of the Theorem

Before presenting the proof, we first provide an outline of the required steps. After the proof, we describe how to generalize it to accommodate non-path-length estimators, such as collision estimators or surface crossing estimators. The outline of the proof is:

3.1 Define functions of random variables f and g for which $F = \frac{1}{\mathrm{vol}(K)}\Big(E[f] + E[g]\Big)$ by completing the following substeps:

- 3.1.1 Formally integrate the characteristic equation for Equation (4a) subject to the boundary condition Equation (4b) over all directions and over the volume enclosed by a single mesh element,
- 3.1.2 Estimate the integral using Monte Carlo:
  - Change variables such that $s = 0$ defines a point on $\partial\mathcal{D}$ instead of $\mathcal{D}$,
  - Define sampling procedures for the volume and boundary sources by defining random variables, a joint probability density function, and a function of the random variables such that the expectation of the function is the integral that we want to compute,

3.2 Define the estimator $\hat{F} \approx F$ as $\hat{F} = \frac{1}{\mathrm{vol}(K)}\left(\frac{1}{N}\sum_{n=1}^{N} f^{(n)} + \frac{1}{M}\sum_{m=1}^{M} g^{(m)}\right)$ where $f^{(n)}$ and $g^{(m)}$ are the nth and mth random variates of f and g, respectively.

3.3 Evaluate the Monte Carlo estimator variance Equation (3) for $\mathrm{Var}[\hat{F}]$ and simplify the expression by applying the TDL scaling Equations (6a) to (6c) to arrive at the function $G(\varepsilon)$ of the exponents of the leading terms in $Q^\varepsilon \sim \varepsilon^{\alpha_Q}$, $\psi_{\mathrm{inc}}^\varepsilon \sim \varepsilon^{\alpha_{\mathrm{inc}}}$, and $u^\varepsilon \sim \varepsilon^{\alpha_u}$ that is specified by the theorem.

The proof is as follows:

Proof. This is the proof of Theorem 1, which is divided into the parts described above.

3.1. Define functions of random variables f and g

We begin by deriving the expectation for the MC estimator $\hat{F}$. The first step in this process is to formally integrate the characteristic equation over all directions and over the volume enclosed by a single mesh element.

3.1.1. Formally integrate the characteristic equation

The characteristic equation for Equation (4a) subject to the boundary condition given by Equation (4b) is,

$$\psi(\boldsymbol{r}, \boldsymbol{\Omega}) = e^{-\int_0^{s_0} \sigma_t(\boldsymbol{r}-\eta\boldsymbol{\Omega})\,\mathrm{d}\eta}\psi_{\mathrm{inc}}(\boldsymbol{r} - s_0\boldsymbol{\Omega}, \boldsymbol{\Omega}) + \int_0^{s_0} e^{-\int_0^{s} \sigma_t(r-\eta\boldsymbol{\Omega})\,\mathrm{d}\eta} Q(\boldsymbol{r} - s\boldsymbol{\Omega}, \boldsymbol{\Omega})\,\mathrm{d}s\,, \tag{24}$$

where we changed notation for the spatial coordinate from $\mathbf{x}$ to $\boldsymbol{r} - s\boldsymbol{\Omega}$, the parameter s is the distance along the characteristic, and s_0 is the distance to the domain boundary in the $-\boldsymbol{\Omega}$ direction. To understand Equation (24), imagine placing a detector at the phase-space point $(\boldsymbol{r}, \boldsymbol{\Omega})$, shown as the black dot at $\boldsymbol{r}$ in Figure 1. The first term in Equation (24) represents particles emitted from the boundary $\partial\mathcal{D}$ in direction $\boldsymbol{\Omega}$ and exponentially attenuated before reaching the detector, while the second term represents particles emitted along s_0 with direction $\boldsymbol{\Omega}$ and similarly attenuated.

Multiply Equation (24) by an arbitrary weighting function $u = u(\boldsymbol{\Omega})$, integrate over the unit sphere, impose a mesh, and represent the quantity described by the angular integral as a piecewise constant function that is single-valued on each mesh element by averaging the angular integral over an arbitrary mesh element K,

$$\begin{aligned}\frac{1}{\mathrm{vol}(K)} \int_K \int_{\mathbb{S}^2} u(\boldsymbol{\Omega})\, \psi(\boldsymbol{r}, \boldsymbol{\Omega})\,\mathrm{d}\Omega\,\mathrm{d}\boldsymbol{r} &= \frac{1}{\mathrm{vol}(K)} \int_K \int_{\mathbb{S}^2} u(\boldsymbol{\Omega})\, e^{-\int_0^{s_0} \sigma_t(\boldsymbol{r}-\eta\boldsymbol{\Omega})\,\mathrm{d}\eta}\psi_{\mathrm{inc}}(\boldsymbol{r} - s_0\boldsymbol{\Omega}, \boldsymbol{\Omega})\,\mathrm{d}\Omega\,\mathrm{d}\boldsymbol{r} \\ &+ \frac{1}{\mathrm{vol}(K)} \int_K \int_{\mathbb{S}^2} \int_0^{s_0} u(\boldsymbol{\Omega})\, e^{-\int_0^{s} \sigma_t(r-\eta\boldsymbol{\Omega})\,\mathrm{d}\eta} Q(\boldsymbol{r} - s\boldsymbol{\Omega}, \boldsymbol{\Omega})\,\mathrm{d}s\,\mathrm{d}\Omega\,\mathrm{d}\boldsymbol{r}\,. \end{aligned} \tag{25}$$

When $u = 1$, Equation (25) is an integral equation for the classic piecewise constant Monte Carlo estimator of the angle integrated intensity. The left-hand side of Equation (25) can be approximated using Monte Carlo integration. If we knew ψ, then the Monte Carlo approximation of Equation (25) could be written as,

$$\frac{1}{\mathrm{vol}(K)} \int_K \int_{\mathbb{S}^2} u(\boldsymbol{\Omega})\, \psi(\boldsymbol{r}, \boldsymbol{\Omega})\,\mathrm{d}\Omega\,\mathrm{d}\boldsymbol{r} \approx \frac{4\pi}{N} \sum_{n=1}^{N} u(\boldsymbol{\Omega}^{(n)})\, \psi(\boldsymbol{r}^{(n)}, \boldsymbol{\Omega}^{(n)})\,, \tag{26}$$

where $\boldsymbol{r}^{(n)}$ and $\boldsymbol{\Omega}^{(n)}$ are random variates of uniform independent and identically distributed (iid) random variables in space and angle, respectively. We do not know ψ, so we use the right-hand side of Equation (25) to compute the sum in Equation (26).

3.1.2. Estimate the integral using Monte Carlo

In this part of the proof, we first change variables such that $s = 0$ defines a point on $\partial\mathcal{D}$ instead of $\mathcal{D}$, then we define sampling procedures for the volume and boundary sources by defining random variables, a joint probability density function, and a function of the random variables such that the expectation of the function is the integral that we want to compute.

Consider the change of variables,

$$\boldsymbol{r}' = \boldsymbol{r} - s\boldsymbol{\Omega}, \qquad \boldsymbol{\Omega}' = \Omega, \qquad s' = s\,. \tag{27}$$

Let $S \sim \mathrm{exponential}(\sigma_t(\boldsymbol{r}' + s\boldsymbol{\Omega}))$ be a random variable representing the distance traveled by a MC photon originating from the point $\boldsymbol{r}'$ and traveling in the direction $\boldsymbol{\Omega}$ before it collides with an electron in the matter. We refer to s as the "path length". The probability density function for the path length s given a fixed position $\boldsymbol{r}'$ and direction $\boldsymbol{\Omega}$ is,

$$p_{\sigma_t}(s) = \sigma_t(\boldsymbol{r}' + s\boldsymbol{\Omega}) e^{-\int_0^{s} \sigma_t(\boldsymbol{r}'+\eta\boldsymbol{\Omega})\,\mathrm{d}\eta}\,. \tag{28}$$

Using the notation $(\cdot\ ; K)$ to emphasize the presence of element K, define the distance traveled in K as,

$$\tau(\boldsymbol{r}', \boldsymbol{\Omega}, s; K) = \begin{cases} 0 & s < s_1(\boldsymbol{r}', \boldsymbol{\Omega}; K)\,, \\ s - s_1(\boldsymbol{r}', \boldsymbol{\Omega}; K) & s_1(\boldsymbol{r}', \boldsymbol{\Omega}; K) \leq s \leq s_2(\boldsymbol{r}', \boldsymbol{\Omega}; K)\,, \\ s_2(\boldsymbol{r}', \boldsymbol{\Omega}; K) - s_1(\boldsymbol{r}', \boldsymbol{\Omega}; K) & s_2(\boldsymbol{r}', \boldsymbol{\Omega}; K) < s\,, \end{cases} \tag{29a}$$

where s_1 and s_2 are the distances to the entry point and exit points of K along $\boldsymbol{\Omega}$, respectively,

$$s_1(\boldsymbol{r}', \boldsymbol{\Omega}; K) = \min\{s \mid \boldsymbol{r}' + s\boldsymbol{\Omega} \in \partial K\}\,, \tag{29b}$$

$$s_2(\boldsymbol{r}', \boldsymbol{\Omega}; K) = \max\{s \mid \boldsymbol{r}' + s\boldsymbol{\Omega} \in \partial K\}\,. \tag{29c}$$

Figure 2 shows $\boldsymbol{r}'$ along with three possible absorption locations corresponding to three values of τ. Observe that $\boldsymbol{r}' \in \partial\mathcal{D}$. The final step in this change of variables $\boldsymbol{r} \to \boldsymbol{r}'$ is to relate the volume elements $\mathrm{d}\boldsymbol{r}$ and $\mathrm{d}\boldsymbol{r}'$.

Let $\mathbf{n}(\boldsymbol{r}')$ denote the surface normal vector at $\boldsymbol{r}' \in \partial\mathcal{D}$. Consider the volume, made by an infinitesimal area element $\mathrm{d}A$ on the boundary $\partial\mathcal{D}$ extruded by distance $\mathrm{d}s$ in direction $\boldsymbol{\Omega}$,

$$\{\boldsymbol{r}' + s\boldsymbol{\Omega} \,|\, \boldsymbol{r}' \in \mathrm{d}A \subset \partial\mathcal{D},\ s \in \hat{s} + \mathrm{d}s\}\,. \tag{30}$$

In two spatial dimensions, this volume is a function of the area of the parallelogram in $\mathrm{span}\{\mathbf{n}, \mathbf{n}^\perp\}$, where $\mathbf{n}^\perp$ is the unit vector perpendicular to the surface normal $\mathbf{n}$, formed from $\mathbf{n} \cdot \mathbf{n}^\perp = 0$ and shown in Figure 3. The volume is,

$$\mathrm{d}s\ \mathrm{d}A\,|\mathbf{n}^\perp \times \boldsymbol{\Omega}| = \mathrm{d}s\ \mathrm{d}A\left|\det\begin{bmatrix} n_2 & \Omega_1 \\ -n_1 & \Omega_2 \end{bmatrix}\right|\,, \tag{31}$$

where the subscripts denote the entry number of the value in the corresponding vector. Equation (31) simplifies to $\mathrm{d}s\ \mathrm{d}A\,|\boldsymbol{\Omega} \cdot \mathbf{n}|$. In three spatial dimensions, the parallelogram is a parallelepiped, and its volume is the triple product of three spanning vectors. Let $\boldsymbol{t}_1$ and $\boldsymbol{t}_2$ span the surface patch and let $\boldsymbol{v} = \boldsymbol{\Omega}\,\mathrm{d}s$ be the extrusion vector. Then the volume is,

$$\mathrm{d}V = |\det\begin{bmatrix}\boldsymbol{t}_1 & \boldsymbol{t}_2 & \boldsymbol{v}\end{bmatrix}| = |(\boldsymbol{t}_1 \times \boldsymbol{t}_2) \cdot \boldsymbol{v}| = |(\boldsymbol{t}_1 \times \boldsymbol{t}_2) \cdot \boldsymbol{\Omega}|\,\mathrm{d}s\,, \tag{32}$$

where $\boldsymbol{t}_1 \times \boldsymbol{t}_2 = \mathbf{n}\,\mathrm{d}A$, so Equation (32) simplifies to $\mathrm{d}V = \mathrm{d}s\ \mathrm{d}A\,|\boldsymbol{\Omega} \cdot \mathbf{n}|$ as before. Also note that,

$$\int_0^s \sigma_t(\boldsymbol{r} - \eta\boldsymbol{\Omega})\,\mathrm{d}\eta = \int_0^s \sigma_t(\boldsymbol{r}' + (s - \eta)\boldsymbol{\Omega})\,\mathrm{d}\eta\,. \tag{33}$$

Let $\tilde{\eta} = s - \eta$, so $\mathrm{d}\tilde{\eta} = -\,\mathrm{d}\eta$. When $\eta = 0$, $\tilde{\eta} = s$, and when $\eta = s$, $\tilde{\eta} = 0$. Thus,

$$\int_0^s \sigma_t(\boldsymbol{r}' + (s - \eta)\boldsymbol{\Omega})\,\mathrm{d}\eta = -\int_s^0 \sigma_t(\boldsymbol{r}' + \tilde{\eta}\boldsymbol{\Omega})\,\mathrm{d}\tilde{\eta}\,. \tag{34}$$

Switching the order of the limits of integration flips the sign,

$$-\int_s^0 \sigma_t(\boldsymbol{r}' + \tilde{\eta}\boldsymbol{\Omega})\,\mathrm{d}\tilde{\eta} = \int_0^s \sigma_t(\boldsymbol{r}' + \tilde{\eta}\boldsymbol{\Omega})\,\mathrm{d}\tilde{\eta}\,, \tag{35}$$

and renaming the dummy variable of integration from $\tilde{\eta}$ to η gives,

$$\int_0^s \sigma_t(\boldsymbol{r}' + \tilde{\eta}\boldsymbol{\Omega})\,\mathrm{d}\tilde{\eta}\,. = \int_0^s \sigma_t(\boldsymbol{r}' + \eta\boldsymbol{\Omega})\,\mathrm{d}\eta\,. \tag{36}$$

Therefore, the integral which is expressed in $\boldsymbol{r}$ coordinates,

$$\int_K e^{-\int_0^{s_0} \sigma_t(\boldsymbol{r} - \eta\boldsymbol{\Omega})\,\mathrm{d}\eta}\psi_{\mathrm{inc}}(\boldsymbol{r} - s_0\boldsymbol{\Omega}, \boldsymbol{\Omega})\,\mathrm{d}\boldsymbol{r}\,, \tag{37}$$

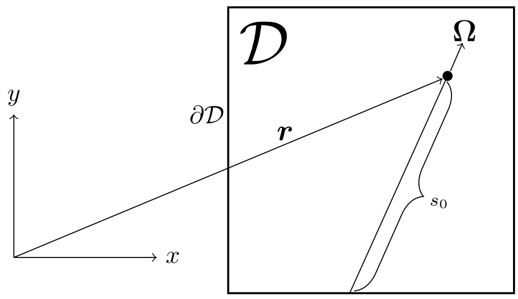


Figure 1: Distance to domain boundary $s_0(\boldsymbol{r})$.

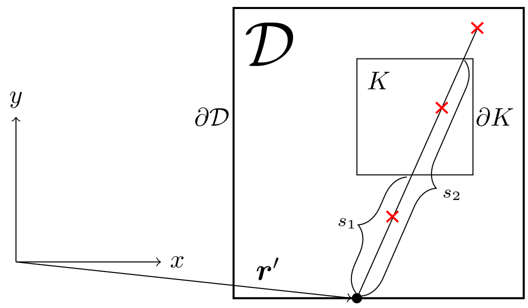
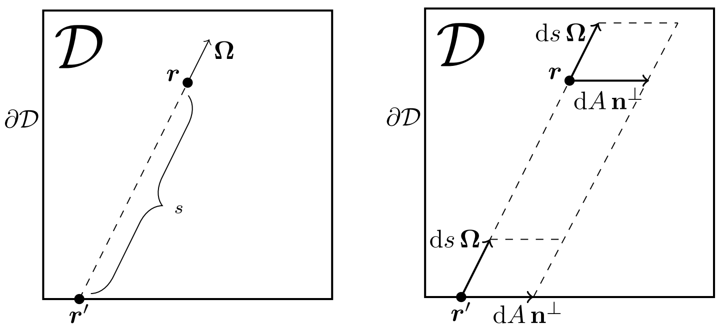


Figure 2: Distance to element entry $s_1(\boldsymbol{r}')$ and exit $s_2(\boldsymbol{r}')$ and three possible absorption locations marked by **X**'s along the path $\boldsymbol{r}' + s\boldsymbol{\Omega}$. Here, $\tau = 0$ for the **X** closest to $\boldsymbol{r}'$, $\tau = s - s_1$ for the middle **X**, and $\tau = s_2 - s_1$ for the far **X**.



(a) Change of variables. (b) Volume scaling.

Figure 3: The change of variables $\boldsymbol{r} \to \boldsymbol{r}'$ generates a factor of $|\boldsymbol{\Omega}\cdot\mathbf{n}|$ in the boundary surface integral because $\mathrm{d}\boldsymbol{r} = |\boldsymbol{\Omega}\cdot\mathbf{n}|\,\mathrm{d}s\,\mathrm{d}A$.

can be equivalently represented as an integral in $\boldsymbol{r}'$ coordinates,

$$\int_K e^{-\int_0^s \sigma_t(\boldsymbol{r}'+\eta\boldsymbol{\Omega})\,\mathrm{d}\eta}\psi_{\text{inc}}(\boldsymbol{r}',\boldsymbol{\Omega})\,|\boldsymbol{\Omega}\cdot\mathbf{n}|\,\mathrm{d}s\,\mathrm{d}A\,. \tag{38}$$

The change of variables $\boldsymbol{r}\to\boldsymbol{r}'$ generates a factor of $|\boldsymbol{\Omega}\cdot\mathbf{n}|$ in the boundary surface integral because the infinitesimal volume element $\mathrm{d}\boldsymbol{r}$ is equal to $|\boldsymbol{\Omega}\cdot\mathbf{n}|\,\mathrm{d}s\,\mathrm{d}A$ in $\boldsymbol{r}'$ coordinates.

We proceed by considering the contribution of the volume source Q and the boundary source ψ_{inc} to $\hat{F}$ separately. First, the volume source.

Volume source contribution to $\hat{F}$

Let $\boldsymbol{R}\sim U(\mathcal{D})$, $\boldsymbol{\omega}\sim U(\mathbb{S}^2)$, and $S\sim\text{exponential}(\sigma_t(\boldsymbol{R}+s\boldsymbol{\omega}))$ be random variables representing the initial position, direction, and distance traveled by a MC photon from the volume source, respectively. Define the joint probability density function of $\boldsymbol{R},\boldsymbol{\omega}$, and S as,

$$p(\boldsymbol{r}',\boldsymbol{\Omega},s)=\frac{1}{\text{vol}(\mathcal{D})}\frac{1}{4\pi}p_{\sigma_t}(\boldsymbol{r}',\boldsymbol{\Omega},s)\,, \tag{39}$$

with domain $\mathcal{A}=\mathcal{D}\times\mathbb{S}^2\times[0,\infty)$ and $p_{\sigma_t}(\boldsymbol{r}',\boldsymbol{\Omega},s)$ defined in Equation (28). Define a function of the random variables $\boldsymbol{R},\boldsymbol{\omega}$, and S representing the contribution of the volume source $Q(\boldsymbol{r}',\boldsymbol{\Omega})$ to the estimator $\hat{F}$,

$$f(\boldsymbol{R},\boldsymbol{\omega},S)=4\pi\ \text{vol}(\mathcal{D})\,u(\boldsymbol{\Omega})\,Q(\boldsymbol{r}',\boldsymbol{\Omega})\,\tau(\boldsymbol{r}',\boldsymbol{\Omega},s;K)\,. \tag{40}$$

If $\psi_{\text{inc}}=0$ then Equation (25) can now be written in terms of the expectation of $f(\boldsymbol{R},\boldsymbol{\omega},S)$,

$$\frac{1}{\text{vol}(K)}E\Big[f(\boldsymbol{R},\boldsymbol{\omega},S)\Big]=\frac{1}{\text{vol}(K)}\int_{\mathcal{D}}\int_{\mathbb{S}^2}\int_0^\infty f(\boldsymbol{r}',\boldsymbol{\Omega},s)\,p(\boldsymbol{r}',\boldsymbol{\Omega},s)\,\mathrm{d}s\,\mathrm{d}\Omega\,\mathrm{d}\boldsymbol{r}'\,, \tag{41}$$

where $p(\boldsymbol{r}',\boldsymbol{\Omega},s)$ is the joint probability density function defined in Equation (39). The Monte Carlo approximation of the expectation in Equation (41) is,

$$E\Big[f(\boldsymbol{R},\boldsymbol{\omega},S)\Big]\approx\frac{1}{N}\sum_{n=1}^N f\left(\boldsymbol{r}^{(n)},\boldsymbol{\Omega}^{(n)},s^{(n)}\right)\,, \tag{42}$$

where N is the number of particles sampled from the volume source and $\left(\boldsymbol{r}^{(n)},\boldsymbol{\Omega}^{(n)},s^{(n)}\right)$ for $n=1,\ldots,N$ are random variates of the random variables $\boldsymbol{R},\boldsymbol{\omega}$, and S. Substituting Equation (40) into Equation (42) gives,

$$\frac{1}{N}\sum_{n=1}^N f\left(\boldsymbol{r}^{(n)},\boldsymbol{\Omega}^{(n)},s^{(n)}\right)=\frac{4\pi\ \text{vol}(\mathcal{D})}{N}\sum_{n=1}^N u\left(\boldsymbol{\Omega}^{(n)}\right)\,Q\left(\boldsymbol{r}^{(n)},\boldsymbol{\Omega}^{(n)}\right)\tau\left(\boldsymbol{r}^{(n)},\boldsymbol{\Omega}^{(n)},s^{(n)};K\right)\,. \tag{43}$$

Equation (43) is the contribution of the volume source to the $\hat{F}$ estimator. The sum in Equation (43) is over all N particles sourced in the volume $\mathcal{D}$. We can rewrite Equation (43) as a sum over all volume source particle paths traversed in some element K. Assume that volume source particle n traversed path i in element K and define its weight as,

$$w_i=\frac{4\pi\ \text{vol}(\mathcal{D})}{N}u\left(\boldsymbol{\Omega}^{(n)}\right)Q\left(\boldsymbol{r}^{(n)},\boldsymbol{\Omega}^{(n)}\right)\,. \tag{44}$$

Define the length of path i as,

$$d_i=\tau\left(\boldsymbol{r}^{(n)},\boldsymbol{\Omega}^{(n)},s^{(n)};K\right)\,. \tag{45}$$

Equation (43) can now be rewritten as the sum over paths traversed by volume source particles in K,

$$\sum_{i=1} w_i d_i\,. \tag{46}$$

Multiplying Equation (46) by the inverse volume in front of the expectation in Equation (41) gives us our final result for the contribution of the volume source to the estimator $\hat{F}$ on K,

$$\frac{1}{\mathrm{vol}(K)} \sum_{i=1} w_i d_i \,. \tag{47}$$

Next, we consider the contribution of the boundary source ψ_{inc} to $\hat{F}$.

Boundary source contribution to $\hat{F}$

Let $\boldsymbol{R}_b \sim U(\partial\mathcal{D})$, $\boldsymbol{\omega}_h \sim U(\mathbb{S}^2_h)$, and $S_b \sim \mathrm{exponential}(\sigma_t(\boldsymbol{R}_b + s\boldsymbol{\omega}_h))$ be random variables representing the initial position, direction, and distance traveled by a MC photon from the boundary source, respectively, where $\mathbb{S}^2_h$ is all directions on the unit hemisphere defined by $\boldsymbol{\Omega}\cdot\mathbf{n} < 0$ at $\mathbf{x} \in \partial\mathcal{D}$. Define the joint probability density function of $\boldsymbol{R}_b, \boldsymbol{\omega}_h$, and S_b as,

$$\rho(\boldsymbol{r}', \boldsymbol{\Omega}, s) = \frac{1}{\mathrm{area}(\partial\mathcal{D})} \frac{1}{2\pi} p_{\sigma_t}(\boldsymbol{r}', \boldsymbol{\Omega}, s) \,, \tag{48}$$

with domain $\mathcal{B} = \partial\mathcal{D} \times \mathbb{S}^2_h \times [0, \infty)$ and $p_{\sigma_t}(\boldsymbol{r}', \boldsymbol{\Omega}, s)$ defined in Equation (28). Define a function $g(\boldsymbol{R}_b, \boldsymbol{\omega}_h, S_b)$ of the random variables $\boldsymbol{R}_b, \boldsymbol{\omega}_h$, and S_b which represents the contribution of the boundary source $\psi_{\mathrm{inc}}(\boldsymbol{r}', \boldsymbol{\Omega})$ to the estimator $\hat{F}$,

$$g(\boldsymbol{R}_b, \boldsymbol{\omega}_h, S) = 2\pi \ \mathrm{area}(\partial\mathcal{D})\, u(\boldsymbol{\Omega})\, \psi_{\mathrm{inc}}(\boldsymbol{r}', \boldsymbol{\Omega})\, \tau(\boldsymbol{r}', \boldsymbol{\Omega}, s; K) \,. \tag{49}$$

If $Q = 0$ then Equation (25) can now be written in terms of the expectation of $g(\boldsymbol{R}_b, \boldsymbol{\omega}_h, S_b)$,

$$\frac{1}{\mathrm{vol}(K)} E\Big[g(\boldsymbol{R}_b, \boldsymbol{\omega}_h, S)\Big] = \frac{1}{\mathrm{vol}(K)} \int_{\partial\mathcal{D}} \int_{\boldsymbol{\Omega}\cdot\mathbf{n}<0} \int_0^\infty |\boldsymbol{\Omega}\cdot\mathbf{n}|\, g(\boldsymbol{r}', \boldsymbol{\Omega}, s)\, \rho(\boldsymbol{r}', \boldsymbol{\Omega}, s)\, \mathrm{d}s\, \mathrm{d}\Omega\, \mathrm{d}A \,, \tag{50}$$

where $\rho(\boldsymbol{r}', \boldsymbol{\Omega}, s)$ is the joint probability density function defined in Equation (48). The Monte Carlo approximation of the expectation in Equation (50) is,

$$E\Big[g(\boldsymbol{R}_b, \boldsymbol{\omega}_h, S)\Big] \approx \frac{1}{M} \sum_{m=1}^{M} g\left(\boldsymbol{r}_b^{(m)}, \boldsymbol{\Omega}_h^{(m)}, s_b^{(m)}\right) \,, \tag{51}$$

where M is the number of particles sampled from the boundary source and $\left(\boldsymbol{r}_b^{(m)}, \boldsymbol{\Omega}_h^{(m)}, s_b^{(m)}\right)$ for $m = 1, \ldots, M$ are random variates of the random variables $\boldsymbol{R}_b, \boldsymbol{\omega}_h$, and S_b. Substituting Equation (49) into Equation (51) gives,

$$\begin{aligned} &\frac{1}{M} \sum_{m=1}^{M} g\left(\boldsymbol{r}_b^{(m)}, \boldsymbol{\Omega}_h^{(m)}, s_b^{(m)}\right) \\ &\quad = \frac{2\pi \ \mathrm{area}(\partial\mathcal{D})}{M} \sum_{m=1}^{M} |\boldsymbol{\Omega}_h^{(m)} \cdot \mathbf{n}|\, u\left(\boldsymbol{\Omega}_h^{(m)}\right)\, \psi_{\mathrm{inc}}\left(\boldsymbol{r}_b^{(m)}, \boldsymbol{\Omega}_h^{(m)}\right) \tau\left(\boldsymbol{r}_b^{(m)}, \boldsymbol{\Omega}_h^{(m)}, s_b^{(m)}; K\right) \,. \end{aligned} \tag{52}$$

Equation (52) is the contribution of the boundary source to the $\hat{F}$ estimator. The sum in Equation (52) is over all M particles sourced on the surface $\partial\mathcal{D}$. We can rewrite Equation (52) as a sum over all boundary source particle paths traversed in some element K. Assume that boundary source particle m traversed path j in element K and define its weight as,

$$w_j = \frac{2\pi \ \mathrm{area}(\partial\mathcal{D})}{M} |\boldsymbol{\Omega}_h^{(m)} \cdot \mathbf{n}|\, u\left(\boldsymbol{\Omega}_h^{(m)}\right)\, \psi_{\mathrm{inc}}\left(\boldsymbol{r}_b^{(m)}, \boldsymbol{\Omega}_h^{(m)}\right) \,. \tag{53}$$

Define the length of path j as,

$$d_j = \tau\left(\boldsymbol{r}_b^{(m)}, \boldsymbol{\Omega}_h^{(m)}, s_b^{(m)}; K\right) \,. \tag{54}$$

Equation (52) can now be rewritten as the sum over paths traversed by boundary source particles in K,

$$\sum_{j=1} w_j d_j \,. \tag{55}$$

Multiplying Equation (55) by the inverse volume in front of the expectation in Equation (50) gives us our final result for the contribution of the boundary source to the estimator $\hat{F}$ on K,

$$\frac{1}{\mathrm{vol}(K)} \sum_{j=1} w_j d_j \,. \tag{56}$$

3.2. Define the Monte Carlo estimator $\hat{F}$

Our choice of f and g allow us to write F as,

$$F = \frac{1}{\mathrm{vol}(K)} \Big(E[f] + E[g] \Big) \,. \tag{57}$$

The estimator $\hat{F}$, which can be computed by taking the sum of Equation (47) and Equation (56), is equivalent to,

$$\hat{F} = \frac{1}{\mathrm{vol}(K)} \left(\frac{1}{N} \sum_{n=1}^{N} f^{(n)} + \frac{1}{M} \sum_{m=1}^{M} g^{(m)} \right) , \tag{58}$$

where $f^{(n)}$ and $g^{(m)}$ are the nth and mth random variates of f and g, respectively. We will use this definition for $\hat{F}$ in terms of f and g to evaluate the Monte Carlo estimator variance Equation (3) for $\mathrm{Var}[\hat{F}]$.

3.3. Evaluate $\mathrm{Var}[\hat{F}]$ and apply the TDL scaling

The variance of the Monte Carlo estimator $\hat{F}$ is the quotient defined in Equation (3) and rewritten here,

$$\mathrm{Var}[\hat{F}] = \frac{\mathrm{Var}[f]}{N} \,. \tag{59}$$

Substituting Equation (58) into Equation (59) gives,

$$\mathrm{Var}[\hat{F}] = \frac{1}{\mathrm{vol}(K)} \left(\frac{\mathrm{Var}[f]}{N} + \frac{\mathrm{Var}[g]}{M} \right) , \tag{60}$$

where we used the fact that the $f^{(n)}$ are independent and identically distributed (iid), the $g^{(m)}$ are iid, and $\mathrm{Cov}[f, g] = 0$ because f and g are independent because they are functions of different independent random variables. To compute $\mathrm{Var}[f]$ and $\mathrm{Var}[g]$, we use the equation for the variance of a function of random variables, which is the expectation of the square minus the square of the expectation, as shown in Equation (1) and rewritten here,

$$\mathrm{Var}[f] = E[f^2] - (E[f])^2 \,. \tag{61}$$

We proceed by considering $\mathrm{Var}[f]$ and $\mathrm{Var}[g]$ separately. First, $\mathrm{Var}[f]$.

Volume source contribution to $\mathrm{Var}[\hat{F}]$

Recalling Equation (41), which shows the expectation of f along with a volume factor, the expectation of f^2 is,

$$E[f^2] = \int_{\mathcal{D}} \int_{\mathbb{S}^2} \int_0^{\infty} f^2 p \, \mathrm{d}s \, \mathrm{d}\Omega \, \mathrm{d}\boldsymbol{r}' \,. \tag{62}$$

Considering only the integral along the particle path, we may substitute Equations (39) and (40) for p and f, respectively, into Equation (62) and simplify to get,

$$\int_0^{\infty} f^2 p \, \mathrm{d}s = 4\pi \ \mathrm{vol}(\mathcal{D}) \, u^2 \, Q^2 \int_0^{\infty} \tau^2 p_{\sigma_t} \, \mathrm{d}s \,, \tag{63}$$

where p_{σ_t} and τ are defined by Equations (28) and (29a), respectively. In a single material problem, the rate in the exponential probability density function is constant, and Equation (28) simplifies to,

$$p_{\sigma_t}(s) = \sigma_t e^{-\sigma_t s} \,. \tag{64}$$

Equation (64) in the TDL is,

$$p_{\sigma_t}(s) = \frac{\sigma_t}{\varepsilon} e^{-\frac{\sigma_t}{\varepsilon} s} \,, \tag{65}$$

where ε is the TDL scaling parameter. Now consider just the integral in Equation (63) without its coefficient. Substituting Equation (65) for p_{σ_t} and Equation (29a) for τ into the integral, and splitting the integral over $s \in [0, \infty)$ to three integrals over $[0, s_1)$, $[s_1, s_2)$, and $[s_2, \infty)$ gives,

$$\int_0^\infty \tau^2 p_{\sigma_t} \, \mathrm{d}s = \int_{s_1}^{s_2} (s - s_1)^2 \frac{\sigma_t}{\varepsilon} e^{-\frac{\sigma_t}{\varepsilon} s} \, \mathrm{d}s + (s_2 - s_1)^2 \int_{s_2}^\infty \frac{\sigma_t}{\varepsilon} e^{-\frac{\sigma_t}{\varepsilon} s} \, \mathrm{d}s \,. \tag{66}$$

The second integral on the right-hand side of Equation (66) is,

$$(s_2 - s_1)^2 \int_{s_2}^\infty \frac{\sigma_t}{\varepsilon} e^{-\frac{\sigma_t}{\varepsilon} s} \, \mathrm{d}s = (s_2 - s_1)^2 e^{-\frac{\sigma_t}{\varepsilon} s_2} \,. \tag{67}$$

The first integral on the right-hand side of Equation (66) may be computed using integration by parts twice. The result is,

$$\begin{aligned} \int_{s_1}^{s_2} (s - s_1)^2 \frac{\sigma_t}{\varepsilon} e^{-\frac{\sigma_t}{\varepsilon} s} \, \mathrm{d}s = 2 \left(\frac{\varepsilon}{\sigma_t} \right)^2 e^{-\frac{\sigma_t}{\varepsilon} s_1} & \\ + \left\{ -(s_2 - s_1)^2 + 2 s_1 \left(\frac{\varepsilon}{\sigma_t} \right) - 2 s_2 \left(\frac{\varepsilon}{\sigma_t} \right) - 2 \left(\frac{\varepsilon}{\sigma_t} \right)^2 \right\} & e^{-\frac{\sigma_t}{\varepsilon} s_2} \,. \end{aligned} \tag{68}$$

We can now write Equation (66) as the sum of Equations (67) and (68),

$$\int_0^\infty \tau^2 p_{\sigma_t} \, \mathrm{d}s = 2 \left(\frac{\varepsilon}{\sigma_t} \right)^2 e^{-\frac{\sigma_t}{\varepsilon} s_1} + \left\{ 2 s_1 \left(\frac{\varepsilon}{\sigma_t} \right) - 2 s_2 \left(\frac{\varepsilon}{\sigma_t} \right) - 2 \left(\frac{\varepsilon}{\sigma_t} \right)^2 \right\} e^{-\frac{\sigma_t}{\varepsilon} s_2} \,. \tag{69}$$

Define a new random variable $\zeta_K^{(1)}$ equal to Equation (69),

$$\zeta_K^{(1)} = 2 \left(\frac{\varepsilon}{\sigma_t} \right)^2 e^{-\frac{\sigma_t}{\varepsilon} s_1} + \left\{ 2 s_1 \left(\frac{\varepsilon}{\sigma_t} \right) - 2 s_2 \left(\frac{\varepsilon}{\sigma_t} \right) - 2 \left(\frac{\varepsilon}{\sigma_t} \right)^2 \right\} e^{-\frac{\sigma_t}{\varepsilon} s_2} \,. \tag{70}$$

We can label the terms in Equation (70) with the leading order of ε for each term in the TDL. Recall that in the TDL, $\varepsilon^p < \varepsilon^q$ for $p > q$, because $\varepsilon \in (0, 1]$. The terms in Equation (70) scale as,

$$\underbrace{\underbrace{2 \left(\frac{\varepsilon}{\sigma_t} \right)^2}_{\varepsilon^2} \underbrace{e^{-\frac{\sigma_t}{\varepsilon} s_1}}_{1}}_{\varepsilon^2} + \underbrace{\left(\underbrace{2 s_1 \left(\frac{\varepsilon}{\sigma_t} \right)}_{\varepsilon} - \underbrace{2 s_2 \left(\frac{\varepsilon}{\sigma_t} \right)}_{\varepsilon} - \underbrace{2 \left(\frac{\varepsilon}{\sigma_t} \right)^2}_{\varepsilon^2} \right) \underbrace{e^{-\frac{\sigma_t}{\varepsilon} s_2}}_{1}}_{\varepsilon} , \tag{71}$$

where the exponentials are bounded by unity because $e^{-x} \in (0, 1]$ for $x \geq 0$. Thus, $\zeta_K^{(1)} \sim \varepsilon$.

Rewriting Equation (62) using Equation (63) with Equation (70) substituted and then simplifying gives,

$$E[f^2] = 4\pi \, \mathrm{vol}(\mathcal{D}) \int_{\mathcal{D}} \int_{\mathbb{S}^2} u^2 \, Q^2 \, \zeta_K^{(1)} \, \mathrm{d}\Omega \, \mathrm{d}\boldsymbol{r}' \,. \tag{72}$$

Replacing u and Q in Equation (72) with their TDL scalings $u^\varepsilon \sim \varepsilon^{\alpha_u}$ and $Q^\varepsilon \sim \varepsilon^{\alpha_Q}$ and using our result $\zeta_K^{(1)} \sim \varepsilon$ allows us to label the terms in Equation (72) with the leading order of ε for each term as,

$$\underbrace{\underbrace{4\pi\ \mathrm{vol}(\mathcal{D})}_{1} \underbrace{\int_{\mathcal{D}} \int_{\mathbb{S}^2} \underbrace{(u^\varepsilon)^2}_{\varepsilon^{2\alpha_u}} \underbrace{(Q^\varepsilon)^2}_{\varepsilon^{2\alpha_Q}} \underbrace{\zeta_K^{(1)}}_{\varepsilon} \,\mathrm{d}\Omega\, \mathrm{d}\boldsymbol{r}'}_{\varepsilon^{2(\alpha_u+\alpha_Q)+1}}}_{\varepsilon^{2(\alpha_u+\alpha_Q)+1}} \,. \tag{73}$$

Thus, $E[f^2] \sim \varepsilon^{2(\alpha_u+\alpha_Q)+1}$.

As shown by Equation (61), the variance is the difference of two terms, of which Equation (72) is only the first. The second is $(E[f])^2$, for which we must consider $E[f]$,

$$E[f] = \int_{\mathcal{D}} \int_{\mathbb{S}^2} \int_0^\infty f p \,\mathrm{d}s\, \mathrm{d}\Omega\, \mathrm{d}\boldsymbol{r}' \,. \tag{74}$$

The integral along the particle path is,

$$\int_0^\infty f p \,\mathrm{d}s = u\, Q \int_0^\infty \tau p_{\sigma_t} \,\mathrm{d}s \,. \tag{75}$$

Substituting Equation (65) for p_{σ_t} and Equation (29a) for τ into the integral and splitting the integral into three as in Equation (66) gives,

$$\int_0^\infty \tau p_{\sigma_t} \,\mathrm{d}s = \int_{s_1}^{s_2} (s - s_1) \frac{\sigma_t}{\varepsilon} e^{-\frac{\sigma_t}{\varepsilon} s} \,\mathrm{d}s + (s_2 - s_1) \int_{s_2}^\infty \frac{\sigma_t}{\varepsilon} e^{-\frac{\sigma_t}{\varepsilon} s} \,\mathrm{d}s \,. \tag{76}$$

The second integral in Equation (76) is,

$$(s_2 - s_1) \int_{s_2}^\infty \frac{\sigma_t}{\varepsilon} e^{-\frac{\sigma_t}{\varepsilon} s} \,\mathrm{d}s = (s_2 - s_1) e^{-\frac{\sigma_t}{\varepsilon} s_2} \,. \tag{77}$$

The first integral in Equation (76) may be computed using integration by parts. The result is,

$$\int_{s_1}^{s_2} (s - s_1) \frac{\sigma_t}{\varepsilon} e^{-\frac{\sigma_t}{\varepsilon} s} \,\mathrm{d}s = \left(s_1 - s_2 - \frac{\varepsilon}{\sigma_t} \right) e^{-\frac{\sigma_t}{\varepsilon} s_2} + \frac{\varepsilon}{\sigma_t} e^{-\frac{\sigma_t}{\varepsilon} s_1} \,. \tag{78}$$

We can now rewrite Equation (76) as the sum of Equations (77) and (78),

$$\int_0^\infty \tau p_{\sigma_t} \,\mathrm{d}s = \frac{\varepsilon}{\sigma_t} \left(e^{-\frac{\sigma_t}{\varepsilon} s_1} - e^{-\frac{\sigma_t}{\varepsilon} s_2} \right) \,. \tag{79}$$

Define a new random variable $\zeta_K^{(0)}$ equal to Equation (79),

$$\zeta_K^{(0)} = \frac{\varepsilon}{\sigma_t} \left(e^{-\frac{\sigma_t}{\varepsilon} s_1} - e^{-\frac{\sigma_t}{\varepsilon} s_2} \right) \,. \tag{80}$$

We can label the terms in Equation (80) with the leading order of ε for each term in the TDL. The terms in Equation (80) scale as,

$$\underbrace{\underbrace{\frac{\varepsilon}{\sigma_t}}_{\varepsilon} \left(\underbrace{e^{-\frac{\sigma_t}{\varepsilon} s_1}}_{1} - \underbrace{e^{-\frac{\sigma_t}{\varepsilon} s_2}}_{1} \right)}_{\varepsilon} \,. \tag{81}$$

Thus, $\zeta_K^{(0)} \sim \varepsilon$. Finally, we can rewrite Equation (74) using Equation (75) with Equation (80) substituted. After simplifying, the result is,

$$E[f] = \int_{\mathcal{D}} \int_{\mathbb{S}^2} u\, Q\, \zeta_K^{(0)}\, \mathrm{d}\Omega\, \mathrm{d}\boldsymbol{r}'\,. \tag{82}$$

Replacing u and Q in Equation (82) with their TDL scalings $u^\varepsilon \sim \varepsilon^{\alpha_u}$ and $Q^\varepsilon \sim \varepsilon^{\alpha_Q}$ and using our result $\zeta_K^{(0)} \sim \varepsilon$ allows us to label the terms in Equation (82) with the leading order of ε for each term as,

$$\underbrace{\int_{\mathcal{D}} \int_{\mathbb{S}^2} \underbrace{u^\varepsilon}_{\varepsilon^{\alpha_u}} \underbrace{Q^\varepsilon}_{\varepsilon^{\alpha_Q}} \underbrace{\zeta_K^{(0)}}_{\varepsilon} \mathrm{d}\Omega\, \mathrm{d}\boldsymbol{r}'}_{\varepsilon^{\alpha_u+\alpha_Q+1}}\,. \tag{83}$$

Thus, $E[f] \sim \varepsilon^{\alpha_u+\alpha_Q+1}$. We can now label the terms in the variance Equation (61). The terms in Equation (61) scale as,

$$\mathrm{Var}[f] = \underbrace{\underbrace{E[f^2]}_{\varepsilon^{2(\alpha_u+\alpha_Q)+1}} - \underbrace{(E[f])^2}_{\varepsilon^{2(\alpha_u+\alpha_Q+1)}}}_{\varepsilon^{2(\alpha_u+\alpha_Q)+1}}\,. \tag{84}$$

We now have the Var$[f]$ term in Equation (60). For the Var$[g]$ term, we must consider the boundary source.

Boundary source contribution to Var$[\hat{F}]$

Recall Equation (50), which shows the expectation of g along with a volume factor. The expectation of g^2 is,

$$E[g^2] = \int_{\partial\mathcal{D}} \int_{\boldsymbol{\Omega}\cdot\mathbf{n}<0} \int_0^\infty |\boldsymbol{\Omega}\cdot\mathbf{n}|\, g^2 \rho\, \mathrm{d}s\, \mathrm{d}\Omega\, \mathrm{d}A\,, \tag{85}$$

where ρ and g are defined by Equations (48) and (49), respectively. If we apply the same procedure that we used to determine the leading order of ε for $E[f^2]$ to Equation (85), we find that $E[g^2] \sim \varepsilon^{2(\alpha_u+\alpha_{\mathrm{inc}})+1}$. Similarly, applying the procedure that we used to determine the leading order of ε for $E[f]$ to $E[g]$, we find that $E[g] \sim \varepsilon^{\alpha_u+\alpha_{\mathrm{inc}}+1}$. This means that Var$[g]$ scales as,

$$\mathrm{Var}[g] = \underbrace{\underbrace{E[g^2]}_{\varepsilon^{2(\alpha_u+\alpha_{\mathrm{inc}})+1}} - \underbrace{(E[g])^2}_{\varepsilon^{2(\alpha_u+\alpha_{\mathrm{inc}}+1)}}}_{\varepsilon^{2(\alpha_u+\alpha_{\mathrm{inc}})+1}}\,. \tag{86}$$

We now have both the Var$[f]$ term and the Var$[g]$ term in Equation (60).

Substituting Equations (84) and (86) into Equation (60) gives the final result,

$$\mathrm{Var}[\hat{F}] = \underbrace{\underbrace{\frac{1}{\mathrm{vol}(K)}}_{1} \left(\underbrace{\frac{\mathrm{Var}[f]}{N}}_{\frac{\varepsilon^{2(\alpha_u+\alpha_Q)+1}}{N}} + \underbrace{\frac{\mathrm{Var}[g]}{M}}_{\frac{\varepsilon^{2(\alpha_u+\alpha_{\mathrm{inc}})+1}}{M}} \right)}_{\varepsilon^{2\alpha_u}\left(\frac{\varepsilon^{2\alpha_Q+1}}{N}+\frac{\varepsilon^{2\alpha_{\mathrm{inc}}+1}}{M}\right)}\,. \tag{87}$$

Thus, the Monte Carlo path length estimator of F has a variance Var$[\hat{F}] \le G(\varepsilon)$ where,

$$G(\varepsilon) \sim \varepsilon^{2\alpha_u} \left(\frac{\varepsilon^{2\alpha_Q+1}}{N} + \frac{\varepsilon^{2\alpha_{\mathrm{inc}}+1}}{M} \right) \quad \text{as } \varepsilon \to 0\,. \tag{88}$$

□

Our proof is specific to path length estimators of F. However, it is possible to prove Theorem 1 for collision estimators or surface crossing estimators by modifying the steps related to defining f and g.

One limitation of our proof is the assumption of a single material. The consideration of non-constant opacities would require using the non-constant-rate exponential probability distribution function Equation (28) throughout, instead of replacing it by the constant rate exponential probability distribution function Equation (64). One should find the resulting definition of $G(\varepsilon)$ unchanged, because the order in ε of $\mathrm{Var}[\hat{F}]$ will not change. Another limitation is the assumption that simulation particles can have no more than one collision. Additionally, we consider uniform sampling of directions on the hemisphere of directions pointing into the domain $\mathbb{S}_h$, and we weight the sampled particles by evaluating $|\mathbf{\Omega} \cdot \mathbf{n}|\psi_{\mathrm{inc}}$. Cosine-law sampling is an alternative choice that samples $\mathbf{\Omega}$ non-uniformly from $p_{\cos}(\mathbf{\Omega}) = \frac{|\mathbf{\Omega}\cdot\mathbf{n}|}{\pi}$ and weights the particles with ψ_{inc} only. One could generalize the proof to account for non-uniform direction sampling, in which case one should find the resulting definition of $G(\varepsilon)$ unchanged.

The proof of Corollary 1 is much shorter than the proof of Theorem 1. The proof is:

Proof. The TDL scalings of u_0, $\boldsymbol{u_1}$, and $\mathbf{u_2}$ are the identity function, so all three are independent of ε, $\alpha_u = 0$, and Theorem 1 holds. □

It is not generally true that $\alpha_u = 0$, but when $\alpha_u = 0$, the expression for $G(\varepsilon)$ simplifies so that its order depends only on Q^ε and $\psi^\varepsilon_{\mathrm{inc}}$:

$$G(\varepsilon)|_{\alpha_u=0} \sim \frac{\varepsilon^{2\alpha_Q+1}}{N} + \frac{\varepsilon^{2\alpha_{\mathrm{inc}}+1}}{M} \quad \text{as } \varepsilon \to 0\,. \tag{89}$$

The application that we highlight in the next section is a result that was obtained because of Theorem 1.

4. Application of the Theorem

Hybrid Second Moment (HSM) is a hybrid Monte Carlo-deterministic method for solving the transport Equation (4a) with Q defined by Equation (5) subject to the inflow boundary condition Equation (4b) without simulating scattering events [5]. Instead, the contribution of the scattering source is incorporated into the weight of the simulation particles. This necessitates an iteration to compute successive approximations of the integral in the scattering source term in Equation (5). Rather than using the angle integrated intensity estimator to approximate the integral, the authors solve a system of moment equations deterministically, then use the solution in place of the integral.

HSM avoids the very long Monte Carlo runtimes that occur in the thick diffusion limit, where the mean free path is arbitrarily small and the scattering probability approaches unity. HSM is a hybrid method because it uses Monte Carlo to solve a transport equation, and a deterministic method to solve a system of moment equations. HSM is an iterative method because it runs a Monte Carlo calculation and a deterministic calculation every cycle until the scattering source term, which is lagged, converges.

The Monte Carlo solve in HSM estimates the angle integrated intensity associated with the equation,

$$\mathbf{\Omega} \cdot \nabla\psi + \sigma_t\psi = \frac{\sigma_s}{4\pi}\varphi + q\,, \tag{90}$$

where $\mathbf{\Omega}, \psi, \sigma_t, \sigma_s$, and q were defined in Section 2, and φ is the deterministic solution of the moment system. Equation (90) is subject to the inflow boundary condition Equation (4b). The total source function Q is,

$$Q = \frac{\sigma_s}{4\pi}\varphi + q\,. \tag{91}$$

We can use Theorem 1 to predict how HSM will perform in the thick diffusion limit (TDL). Define $Q^\varepsilon_{\mathrm{HSM}}$ to be Q from Equation (91) subject to the TDL scalings in Equations (6a) to (6c). As we did in the proof of Theorem 1, we can label the terms in $Q^\varepsilon_{\mathrm{HSM}}$ with the leading order of ε for each term in the TDL. The terms in $Q^\varepsilon_{\mathrm{HSM}}$ scale as,

$$Q^\varepsilon_{\mathrm{HSM}} = \underbrace{\frac{\sigma_t/\varepsilon - \sigma_a \varepsilon}{4\pi}\varphi^\varepsilon}_{\varepsilon^{-1}} + \underbrace{q\varepsilon}_{\varepsilon}\,, \tag{92}$$

where we used $\varphi^\varepsilon \sim 1$. Thus, $Q^\varepsilon_{\mathrm{HSM}} \sim \varepsilon^{-1}$ since $\varepsilon^{-1} \gg \varepsilon$ for $\varepsilon \in (0,1] \to 0$. By Theorem 1, $\mathrm{Var}[\cdot] \leq G(\varepsilon) \sim \varepsilon^{-1}$ for path length estimators computed in a weighted Monte Carlo particle simulation of the transport problem defined by Equation (90) subject to the boundary condition Equation (4b), assuming that $\psi^\varepsilon_{\mathrm{inc}} \sim 1$. The variance of Monte Carlo estimators used in the HSM method is unbounded in the TDL, which caused a significant noise issue in HSM calculations.

The authors fix the noise issue by fixing the undesirable dependence of $G(\varepsilon)$ on the TDL parameter ε. They do this by substituting $\psi = \bar{\varphi}/(4\pi) + \tilde{\psi}$ into Equation (90), where $\bar{\varphi}$ is an arbitrary function which they define to satisfy certain properties, including that $(\varphi^\varepsilon - \bar{\varphi}^\varepsilon) \sim \varepsilon$. The result is,

$$\mathbf{\Omega} \cdot \nabla \tilde{\psi} + \sigma_t \tilde{\psi} = \frac{\sigma_t}{4\pi}(\varphi - \bar{\varphi}) - \frac{1}{4\pi}(\sigma_a \varphi + \mathbf{\Omega} \cdot \nabla \bar{\varphi}) + q\,. \tag{93a}$$

They substitute $\psi = \bar{\varphi}/(4\pi) + \tilde{\psi}$ into Equation (4b), which gives a new inflow boundary condition,

$$\tilde{\psi}(\mathbf{x}, \mathbf{\Omega}) = \psi_{\mathrm{inc}}(\mathbf{x}, \mathbf{\Omega}) - \frac{1}{4\pi}\bar{\varphi}(\mathbf{x})\,, \quad \mathbf{x} \in \partial\mathcal{D} \text{ and } \mathbf{\Omega} \cdot \mathbf{n} < 0\,. \tag{93b}$$

We can apply Theorem 1 once more, this time to the new transport problem defined by Equations (93a) and (93b). Define $Q^\varepsilon_{\mathrm{new}}$ to be the total source function in Equation (93a). The terms in $Q^\varepsilon_{\mathrm{new}}$ scale as,

$$Q^\varepsilon_{\mathrm{new}} = \underbrace{\underbrace{\underbrace{\frac{\sigma_t/\varepsilon}{4\pi}}_{\varepsilon^{-1}} \underbrace{(\varphi^\varepsilon - \bar{\varphi}^\varepsilon)}_{\varepsilon}}_{1} - \underbrace{\frac{1}{4\pi}(\underbrace{\sigma_a \varepsilon \varphi^\varepsilon}_{\varepsilon} + \underbrace{\mathbf{\Omega} \cdot \nabla \bar{\varphi}^\varepsilon}_{1})}_{1} + \underbrace{q\varepsilon}_{\varepsilon}}_{1}\,, \tag{94}$$

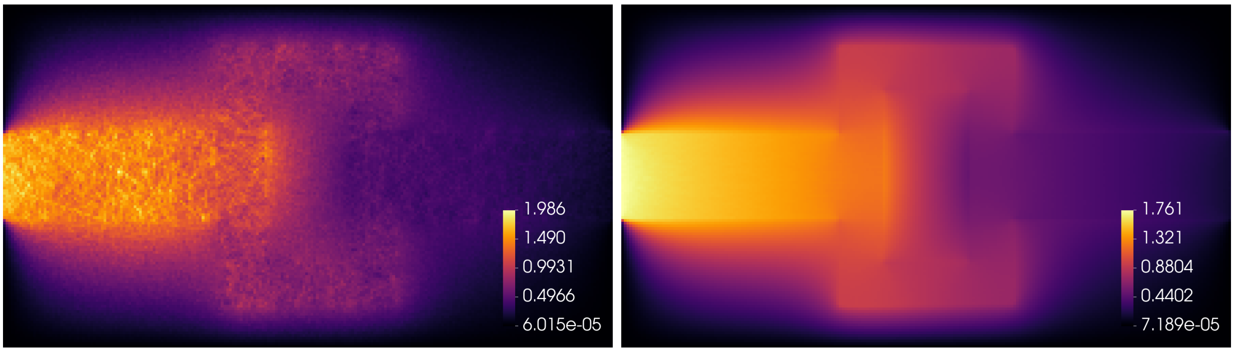
where we used $\varphi^\varepsilon \sim 1$ and $\bar{\varphi}^\varepsilon \sim 1$. The terms in the new inflow boundary condition Equation (93b) scale as,

$$\tilde{\psi}(\mathbf{x}, \mathbf{\Omega}) = \underbrace{\psi^\varepsilon_{\mathrm{inc}}(\mathbf{x}, \mathbf{\Omega})}_{1} - \underbrace{\frac{1}{4\pi}\bar{\varphi}^\varepsilon(\mathbf{x})}_{1}\,, \tag{95}$$

where we have assumed once more that $\psi^\varepsilon_{\mathrm{inc}} \sim 1$. By Theorem 1, $\mathrm{Var}[\cdot] \leq G(\varepsilon) \sim \varepsilon$ for path length estimators computed in a weighted Monte Carlo particle simulation of the transformed problem defined by Equation (93a) subject to the boundary condition Equation (93b).

Thus, by using Theorem 1, the authors of HSM were able to take a method which has a variance that is unbounded in the TDL and create a method which has a variance that limits to zero in the TDL. The variance reduction procedure guided by Theorem 1 allowed the HSM authors to make HSM useful, instead of useless. Figure 4 shows the result of solving a proxy problem from radiative transfer using HSM, both with and without variance reduction, with 16 million particles in each case. In this problem, a light shines on a subset of the left edge of the domain, and photons entering the domain propagate preferentially through an optically-thin region which has an optically-thick obstruction in the center.

The proxy problem emulates the Crooked Pipe, which is a time-dependent radiative transfer problem originally defined in [6]. Figure 4 shows pseudocolor plots of the angle integrated intensity calculated using HSM for the Linearized Crooked Pipe, which is a steady-state version of Crooked Pipe defined in [7]. The linearization of the thermal radiative transfer equations which leads to a linear transport equation for the Linearized Crooked Pipe is equivalent to calculating one very large timestep of a backward difference time integrator. The size of the timestep Δt is such that $c\Delta t = 10^3$, where c is the speed of light in a vacuum. The pipe is 1000 times optically thinner than the surrounding medium.



(a) (b)

Figure 4: Numerical solution before (a) and after (b) applying variance reduction using Theorem 1.

Variance reduction makes the noise nearly imperceptible. By "imperceptible", we simply mean that the pseudocolor plot in Figure 4b appears completely smooth. The pseudocolor plot in Figure 4a exhibits substantial speckling, which is a result of the noise issue: relatively long paths traversed by relatively high-weight particles create bright streaks in the plot. This high-frequency (short-wavelength) spatial variation of the numerical solution is unphysical. The authors found that they needed approximately 500 times more particles (8 billion instead of 16 million) to achieve imperceptible noise without variance reduction. Details for the calculation and the HSM method may be found in [5].

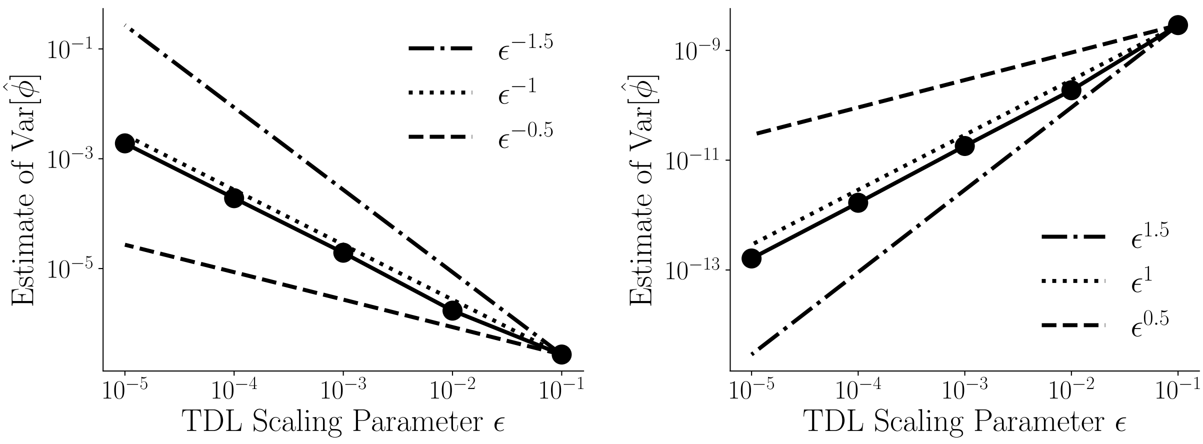
The authors of the same manuscript performed single-zone, single-material calculations to verify that the variance of their path length estimators exhibit the correct order dependence on the TDL parameter ε, as prescribed by Theorem 1. They use five different optical thickness parameter values for the material: $\varepsilon = 10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}$, and 10^{-5}. They estimate the variance of the path length estimator $\hat{\phi}$ for the angle integrated intensity (their estimator is equivalent to the estimator $\hat{F}_0$ that we defined in Corollary 1). Their domain is $\mathcal{D} = [0,1]^2$, they use 100 million particles for each calculation, and they run 600 realizations of each calculation using different pseudo-random number generator seeds. They estimate the variance as the sum of squares of the difference of the estimator $\hat{\phi}$ and the mean estimate,

$$\mathrm{Var}[\hat{\phi}] \approx \frac{1}{600} \sum_{i=1}^{600} (\hat{\phi}_i - \Phi)^2 \,, \tag{96}$$

where Φ is the mean estimate defined by $\Phi = (1/600) \sum_{i=1}^{600} \hat{\phi}_i$. Thus, the total number of calculations is $600 \times 5 = 3000$. Figure 5 shows a plot of their results, reproduced with their permission. They observed the prescribed variances, which limit to infinity and to zero as shown in Figure 5 (a) and (b), respectively. The $G(\varepsilon) \sim \varepsilon^{-1}$ and $G(\varepsilon) \sim \varepsilon$ in Figure 5 (a) and (b) correspond to the TDL scalings $Q^{\varepsilon}_{\mathrm{HSM}}$ and $Q^{\varepsilon}_{\mathrm{new}}$ shown in Equation (92) and Equation (94) in this manuscript, respectively.

5. Conclusions

We presented a proof of a theorem describing the variance of Monte Carlo path length estimators in an important asymptotic regime characterized by an optically-thick, highly scattering material. Our proof considers path length estimators for Monte Carlo particle simulation in a single material for simulation particles that have no more than one collision, but we suggested ways in which one could alleviate some of these limitations. The authors of a radiative transfer method used the theorem to develop a variance reduction technique that enabled a Monte Carlo solution estimator to achieve imperceptible noise with approximately 500 times fewer simulation particles than would be required without variance reduction.



(a) Variance of HSM estimator for Equation (90).

(b) Variance of HSM estimator for Equation (93a).

Figure 5: Hybrid Second Moment estimator variance before variance reduction (a) and after variance reduction (b). Reproduced with permission from the authors of Pozulp et al. [5].

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Authors' contributions

CRediT: **Michael Pozulp**: Investigation, Software, Visualization, Writing – original draft; **Terry Haut**: Conceptualization, Formal analysis, Methodology, Writing – review & editing.

Disclosure statement

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